

FAANG-Level Amazon Data Engineer 6-Month Master Plan

Profile: Beginner | 2 hrs/day | Goal: Amazon Data Engineer

This plan is optimized for working professionals with strict prioritization and high ROI topics.

Month 1-2: Python + SQL + DSA Foundations (Weightage: 45%)

Subtopics: - Python: Data types, loops, functions, OOP, file handling - SQL: Joins, window functions, CTEs, indexing - DSA: Arrays, strings, hashmaps, recursion
Daily Plan: Day 1-3: Python basics revision Day 4-7: SQL joins + practice Day 8-14: DSA arrays + strings Repeat cycle weekly with increasing difficulty
Resources: YouTube: <https://www.youtube.com/watch?v=rfscVS0vtbw>
Docs: <https://docs.python.org/3/tutorial/> GitHub: <https://github.com/TheAlgorithms/Python> SQL Practice: <https://leetcode.com/problemset/database/> Tips: - Spend 1 hr coding + 1 hr SQL daily - Focus on medium-level problems

Month 3: Data Engineering Fundamentals (Weightage: 20%)

Subtopics: - ETL vs ELT - Data pipelines - Batch vs Streaming - Airflow DAGs Daily Plan: Day 1-5: ETL concepts Day 6-10: Build pipeline project Day 11-15: Airflow hands-on Resources: YouTube: <https://www.youtube.com/watch?v=AHMm1wfGuHE> Docs: <https://airflow.apache.org/docs/> GitHub: <https://github.com/apache/airflow> Project: Build ETL pipeline using Python + Airflow

Month 4: Big Data (Weightage: 15%)

Subtopics: - Spark Core - Spark SQL - Partitioning Resources: YouTube:

https://www.youtube.com/watch?v=_C8kWso4ne4 Docs: <https://spark.apache.org/docs/latest/>

GitHub: <https://github.com/apache/spark> Tips: - Focus on transformations and actions

Month 5: AWS Cloud (Weightage: 15%)

Subtopics: - S3, Redshift, Glue, Lambda Resources: YouTube:

<https://www.youtube.com/watch?v=ulprqHHWIng> Docs: <https://docs.aws.amazon.com/> GitHub:

<https://github.com/aws-samples> Project: Build cloud pipeline using AWS

Month 6: System Design + Interview Prep (Weightage: 5%)

Subtopics: - Data system design - Mock interviews Resources: YouTube:

https://www.youtube.com/watch?v=UzLMhgg3_Wc Docs:

<https://github.com/donnemartin/system-design-primer> Tips: - Practice explaining systems clearly

Resume & Portfolio Strategy

- Highlight projects (ETL + AWS) - Use metrics (processed X GB data) - Add GitHub links - Align with Amazon Leadership Principles